Nicholas Oday

CONTEXT OF THE TEACHING / LEARNING SITUATION

Course - Digital Logic Gates
Students - 6 students
Duration - Spring Semester
Schedule - Twice per week, 2 hour sessions
Course Delivery - In Person
Lecture Format - Lecture + Labs

The physical elements of the learning environment that may affect the class include:

- Projectors for demonstration
- Physical breadboards and Integrated circuits
- Proteus, a digital circuit simulator

GENERAL CONTEXT OF THE LEARNING SITUATION

The university expects the course to help prepare students for advanced courses in their major

NATURE OF THE SUBJECT

The course is a mix of theory and hands-on practice, with a 40% theory / 60% hands-on application

The course is primarily convergent as correct answers exists with the use of truth tables, but there is room for creative circuit design

CHARACTERISTICS OF THE LEARNERS

Level: The students are first-year engineering students in their first or second semester

Prior knowledge: The students should have basic understanding of mathematical principles like addition, subtraction, etc. The students are also not required to have prior electronics or programming experience

Student Motivation: This course is required for their major

Some students might have preconceived notions of ~~co~~ science, and the course will challenge them

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CHARACTERISTICS OF THE TEACHER

- I have a strong technical expertise in embedded systems and circuit designs
- I am good at Proteus software
- I have been privately tutoring for 10 years
- My teaching philosophy is:
  - Participation based and collaborative learning
  - Low barrier to entry, or no barrier to entry
  - I want to be a facilitator, a collaborator, and not a dictator

LEARNING GOALS

1 FOUNDATIONAL KNOWLEDGE

- I want the students to understand basic Boolean algebra
- I want students to know and understand truth tables for AND, OR, NOT, NAND gates
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2 APPLICATION

- I want students to be able to design digital circuits using logic gates
- I want students to creatively simplify Boolean expressions using Boolean algebra or truth tables
- I want students to prototype and build circuits on a breadboard using proteus

3 INTEGRATION

- I want students to easily connect the concept of logic gates to everyday circuitry they see, and understand how computers think
- [illegible]
- I want students to see how simple gates combine to create complex circuits

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4 HUMAN DIMENSIONS

- I want students with prior conceptions about engineering and STEM to challenge that conception and discover that they have a place in engineering
- I want students to learn to collaborate well with their fellow students

5 CARING

- I want students to learn and develop an interest for how computers think
- I want students to develop a problem solving and collaborative mindset

6 LEARNING HOW TO LEARN

- I want students to learn to ask questions from me the teacher and their fellow students
- I want students to learn how to debug issues using proven